REVIEW 0 | PROTOTYPE + MATLAB DEMO

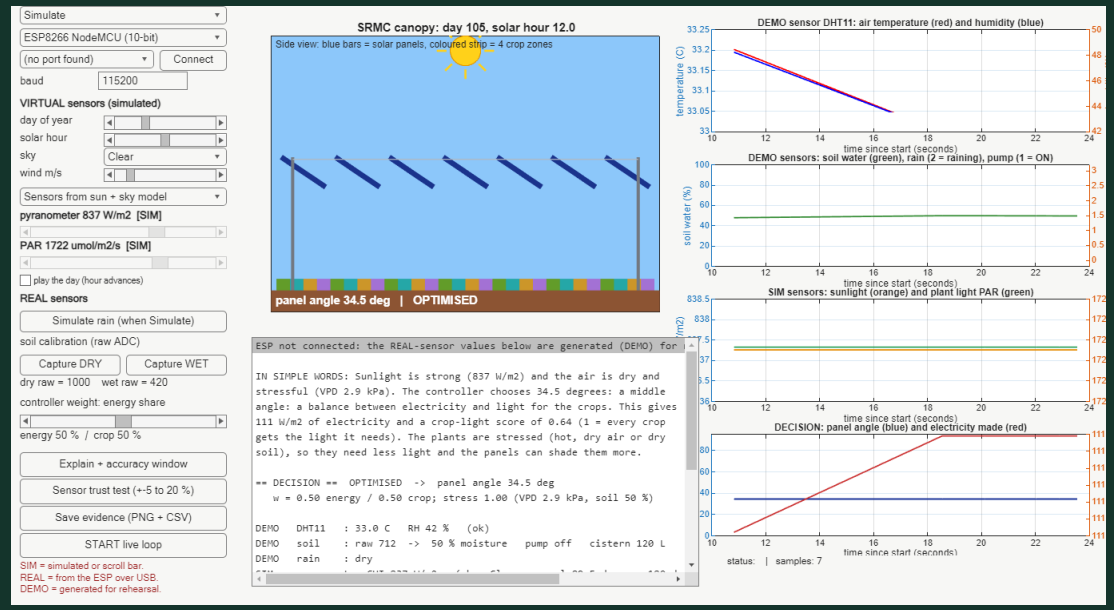
Smart tilt for solar panels over crops: more electricity, and the plants still get their light

Live sensors on an ESP board + a MATLAB dashboard that decides the panel angle, the irrigation and the MPPT.

Site: TCE ground, Madurai (9.88 N).

REAL: DHT11, soil, rain | **SIM: sunlight + plant light** | **MPPT included**

REAL = measured by the sensors on the prototype. SIM = simulated, set with scroll bars in the dashboard (no pyranometer / PAR sensor yet).



The MATLAB dashboard (clear noon): canopy drawing, live plots, decision in words

Student: _____

Guide: _____ Date: _____

How the system works, step by step

From the sensor to the decision, in five steps

1 Sense
DHT11, soil probe and rain sensor read the field

2 Send
ESP board sends one line per second over USB

3 Decide
MATLAB picks the best panel angle (rules + search)

4 Act
Panel angle, irrigation pump and MPPT voltage

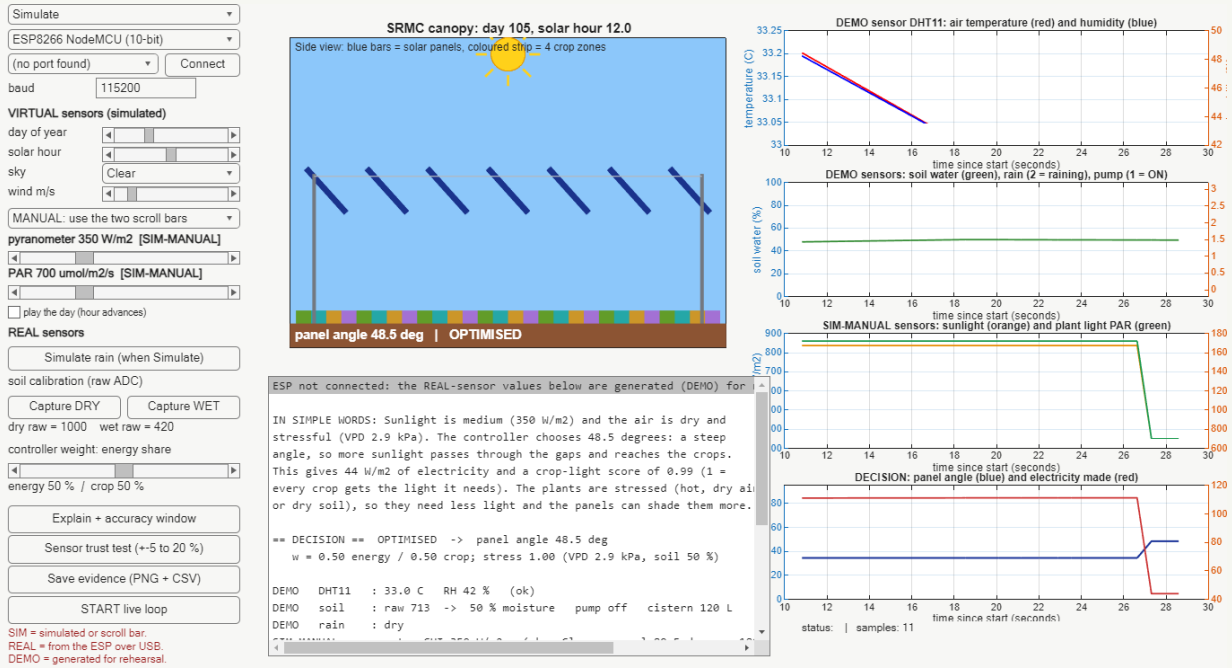
5 Show
Dashboard shows numbers, graphs and the reason in words

Input	Where it comes from	Status	Proof / how
Air temperature, humidity	DHT11 on the ESP board	REAL	datasheet accuracy: 2 C and 5 % RH; checked against a thermometer on the day
Soil moisture	soil probe -> analog pin	REAL	two-point calibration: dry in air = 0 %, wet in water = 100 %
Rain	rain sensor module	REAL	wet plate gives rain = 1; sets the 3 degree slope for harvesting
Sunlight (pyranometer)	scroll bar, or sun model	SIM	sun position from date, time and latitude; clearness index of TCE from NASA POWER
Plant light (PAR)	scroll bar, or 0.45 x 4.57 x sunlight	SIM	about 45 % of sunlight is PAR; 4.57 micromol per joule (standard conversion)
Wind	scroll bar	SIM	safety rule: 15 m/s or more turns the panels edge-on (90 degrees)

Real sensor values on the day: temperature _____ C humidity _____ % soil _____ % rain _____ (fill from the serial monitor / dashboard)

The two simulated sensors: scroll bars, and can we trust them?

Move the scroll bar and watch the angle, the electricity and the irrigation react

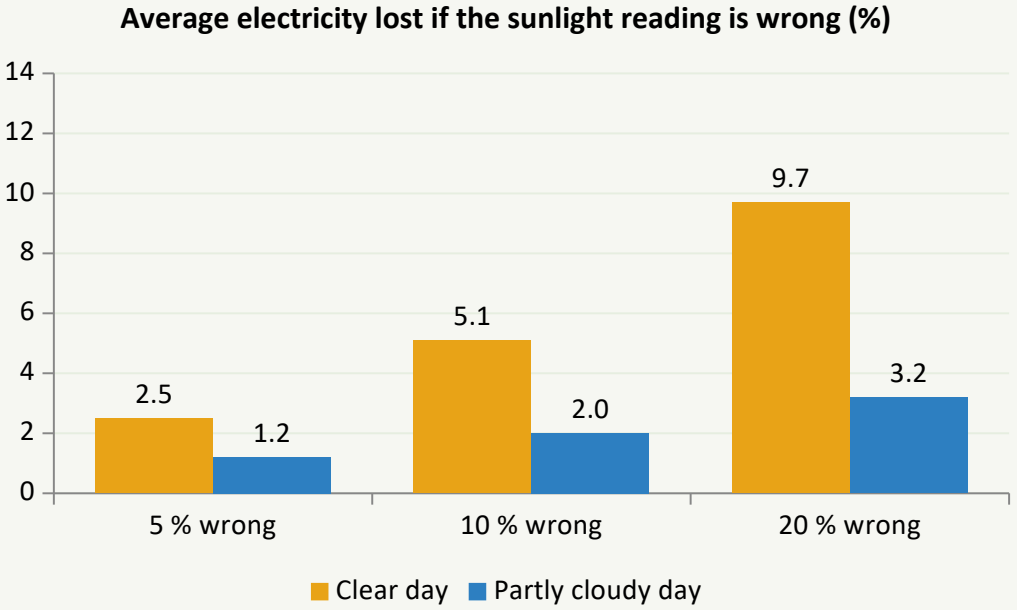


Manual mode: sunlight 350 W/m2, PAR 700 micromol/m2/s

PROOF: how it works and where it comes from

- Sun position: standard solar equations; sky clearness for TCE: NASA POWER monthly data
- PAR = 0.45 x 4.57 x sunlight (standard conversion); the dashboard flags it if PAR / sunlight is far from 2.06
- Same equations were checked against the PVGIS service: +5.0 % per year

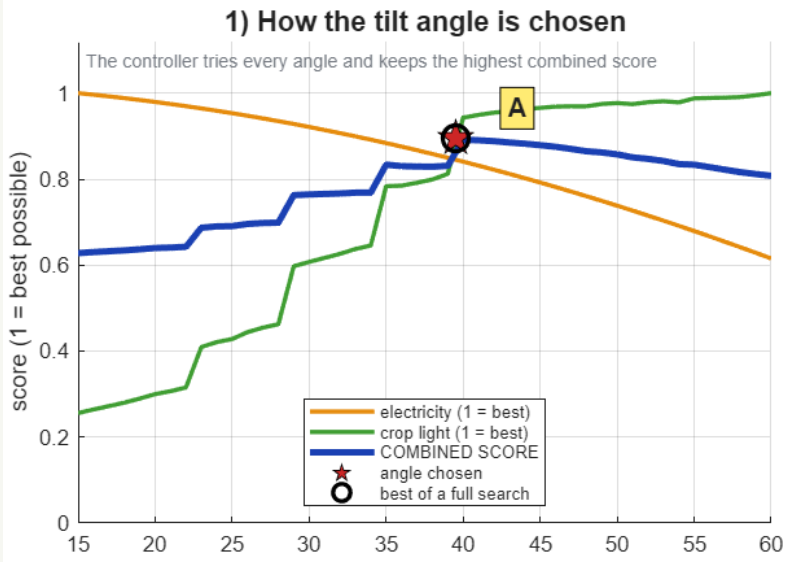
- **Dim sky (300 W/m2):** panels go steep (55.5 degrees) to give crops light; electricity 36 W/m2
- **Bright sky (1000 W/m2):** flatter (39.5 degrees); electricity 124 W/m2
- **Real sensors later:** the serial line takes two extra numbers and replaces the simulated ones

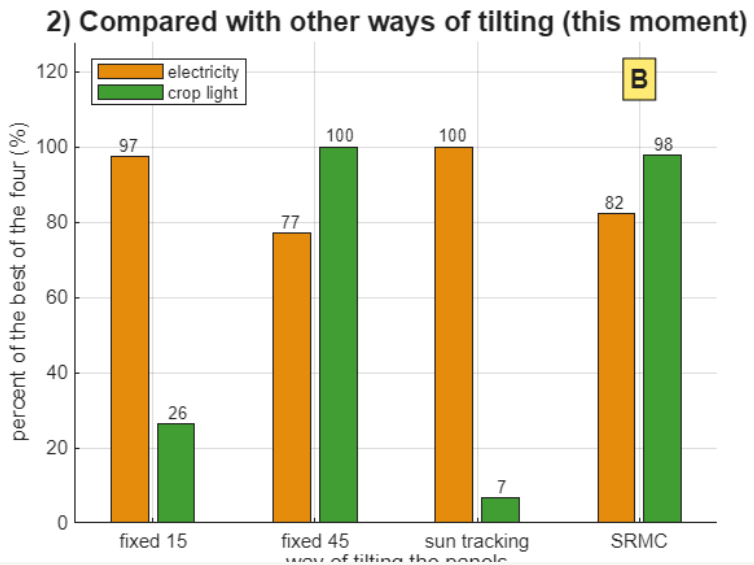


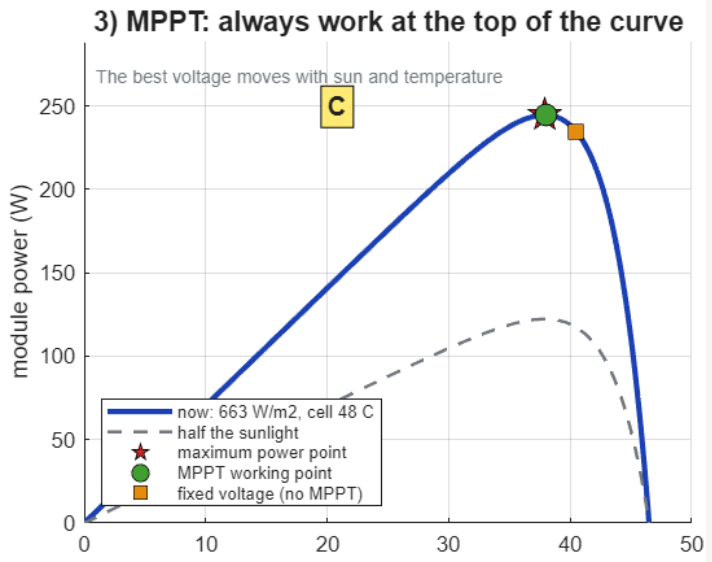
How the controller chooses the tilt angle

It tries every angle from 15 to 60 degrees and keeps the one with the best combined score

How the SRMC system decides the panel angle, and where its accuracy comes from







Rules first (safety and water)

- Wind 15 m/s or more: panels edge-on (90 degrees)
- Rain detected: 3 degree slope, rain goes to the tank
- Night: rest at 15 degrees

Then the search (graph 1)

- Amber line: electricity at each angle
- Green line: light that reaches the crops
- Blue line: combined score = 0.5 x electricity + 0.5 x crop light; red star = the angle chosen

Stress changes the need

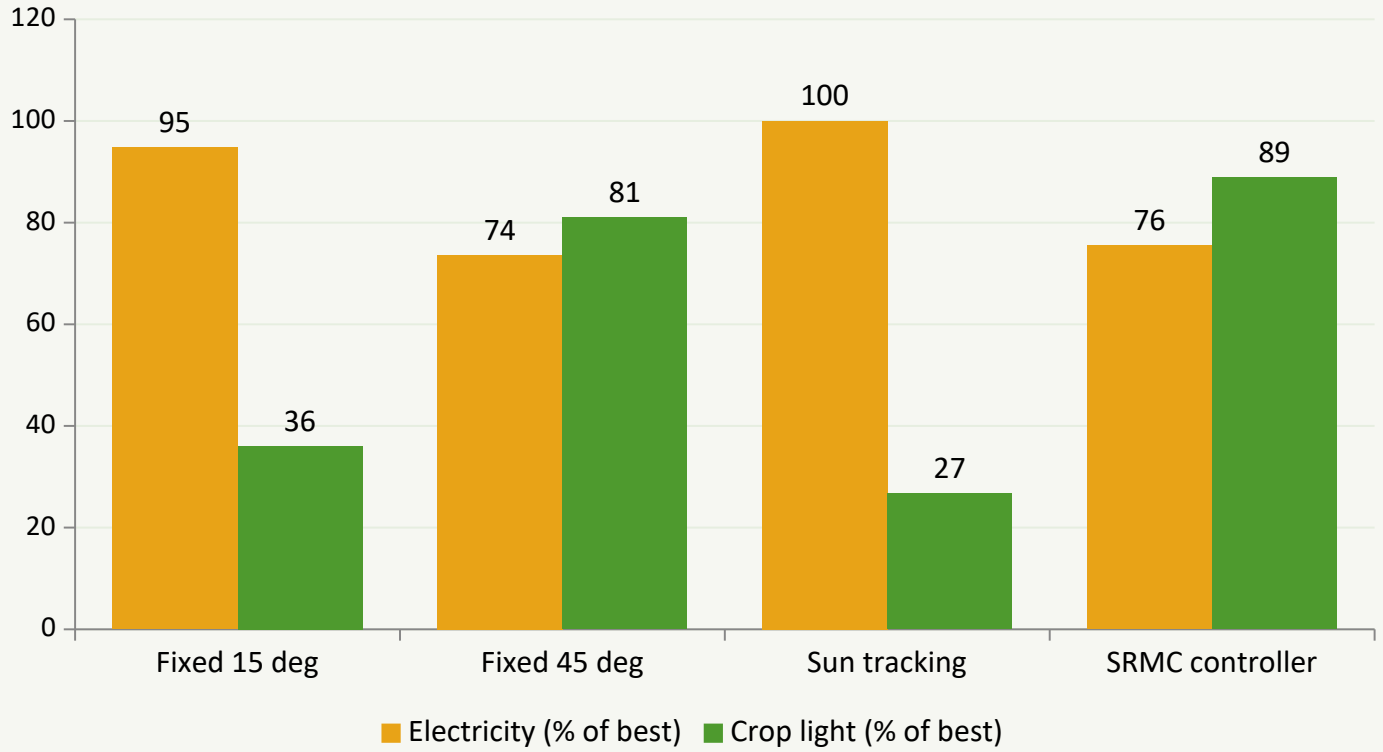
- Hot, dry air (VPD above 2.5 kPa) or dry soil: crops need less light
- The panels can then shade more and make more electricity
- Dry soil: pump on below 35 %, off at 55 %

PROOF

- Accuracy point A: the chosen angle is compared with a full search of every 0.5 degree step. Clear noon: difference 0.0 degrees. Worst case tested: 3 degrees, with the score within 0.02 % (a flat top). Light and shadow come from the same equations checked against Blender 3-D ray tracing (mean difference 0.010).

Result: electricity and crop light together (clear day)

Each bar is a percentage of the best way for that quantity; simulated day, 6 to 18 h



Other days (same controller)	Electricity	Crop light
Partly cloudy day	78 % of best	97 % of best
Hot, dry day (plant stress)	79 % of best	88 % of best

In simple words

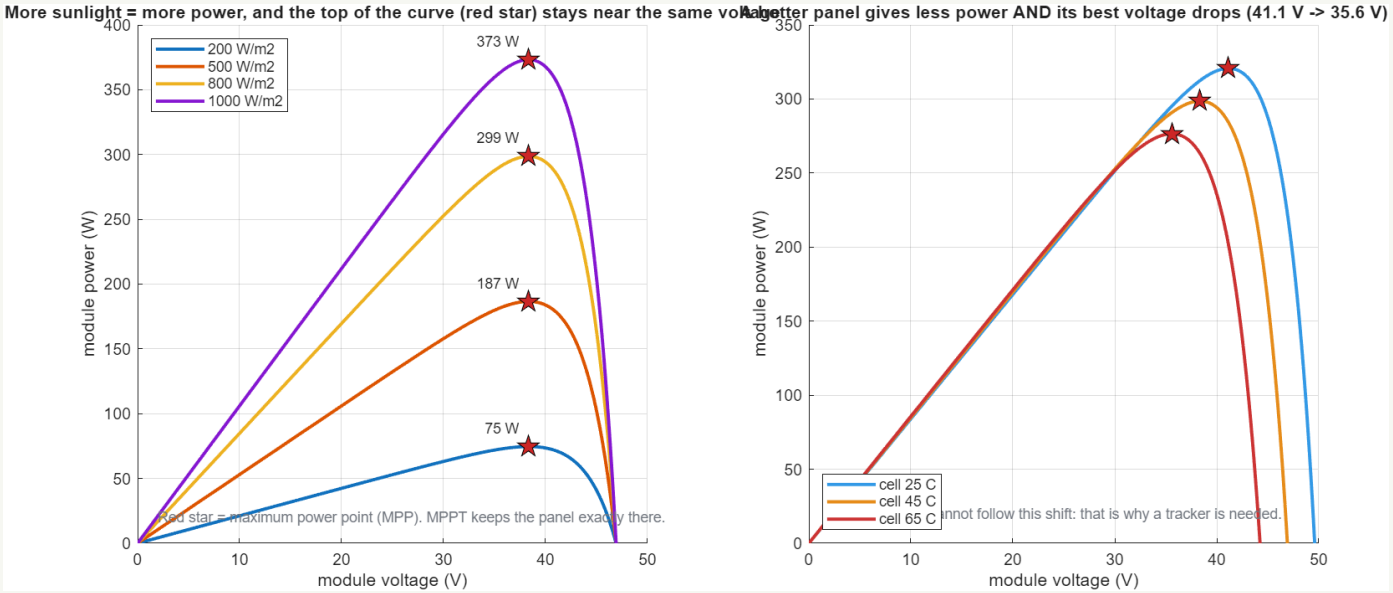
- **Sun tracking:** most electricity, but the crops get only 27 % of the light they need.
- **Fixed 45 degrees:** crops fine, electricity 74 % of best.
- **SRMC controller:** 75 % of the best electricity and 89 % of the best crop light.
- **Against fixed 45 degrees:** electricity +2.5 %, crop light +9.7 %.

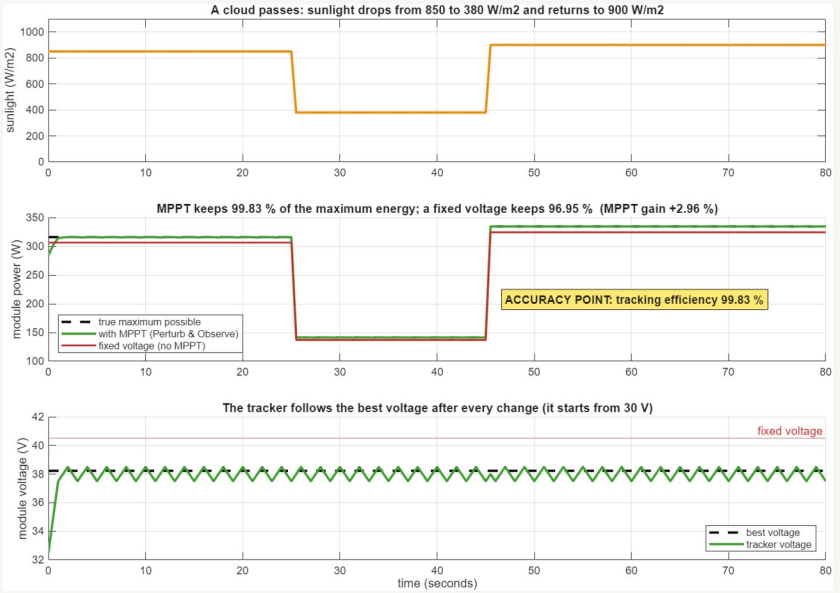
PROOF: how it works and where it comes from

- Simulated day 105, TCE; same functions as the dashboard
- Sun tracking is only a reference: our panels have two positions
- Files: results/day_comparison.csv, fig_energy_vs_crop_clear_day.png

MPPT: always working at the top of the power curve

Maximum power point tracking, Perturb and Observe





What it is

A solar panel gives the most power at one voltage (red star). That voltage moves with sunlight and cell temperature.

What MPPT does

A small controller nudges the voltage up and down and keeps the point where power is highest.

What we gain

Cloud test: MPPT keeps 99.83 % of the maximum energy; a fixed voltage keeps 96.95 % (gain +2.96 %).

PROOF

- Model: single-diode PV model of a typical 400 W module (Isc 10.5 A, Voc 49.6 V, Vmp 40.5 V; replace with your datasheet). Tracker: Perturb and Observe, 0.5 V steps. Hotter cell: best voltage falls from 41.1 V (25 C) to 35.6 V (65 C).
- Accuracy point C: tracking efficiency 99.83 %. Tested in MATLAB (5 MPPT tests pass). This is a model of the tracker, not yet a hardware DC-DC converter.

Accuracy points and proof

Every number: what it means, and where it comes from

	Accuracy point	Number	Where it comes from
A	Angle chosen vs full 0.5 degree search	0.0 degrees at clear noon (worst tested 3 degrees, score within 0.02 %)	search built into the dashboard (Explain window)
B	Gain over a fixed 45 degree panel	electricity +2.5 %, crop light +9.7 % (clear day)	results/day_comparison.csv
C	MPPT tracking efficiency	99.83 % (fixed voltage 96.95 %)	results/mppt_study.md
D	Real sensors	DHT11: 2 C, 5 % RH (datasheet). Bench: _____ C vs _____ C	datasheet; thermometer check on the day
E	Solar and shadow equations	+5.0 % per year vs PVGIS; shadow difference 0.010 vs Blender	PVGIS service; Blender Cycles ray tracing
F	Simulated-sensor error	wrong by 5 %: 2.5 % less electricity; by 10 %: 5.1 % (worst half-hour 35 %)	results/sensor_trust_study.csv
T	Software tests of every rule	29 of 29 pass	results/test_results.md, sensor_and_mppt_tests.md

What we do NOT claim

- Sunlight and PAR are simulated, not measured
- No field data yet; the tilt is shown on screen only
- MPPT and module data are for a typical 400 W module

Next steps

- Connect a real pyranometer and PAR sensor
- Add a motor and a hardware MPPT converter
- Log a week of real sensor data and compare